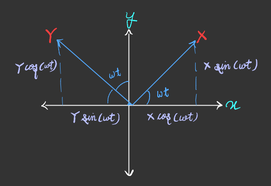
Example 2 : X, Y rotate while x, y are fixed.



$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}_i} = \frac{\partial \mathcal{L}}{\partial x_i}$$

$$X = x\cos(\omega t) + y\sin(\omega t)$$

$$Y = -x\sin(\omega t) + y\cos(\omega t)$$

$$\dot{X} = \left[\dot{x}\cos(\omega t) - x\omega\sin(\omega t)\right] + \left[\dot{y}\sin(\omega t) + y\omega\cos(\omega t)\right]$$

$$\dot{Y} = \left[-\dot{x}\sin(\omega t) - x\omega\cos(\omega t)\right] + \left[\dot{y}\cos(\omega t) - y\omega\sin(\omega t)\right]$$

$$\mathcal{L} = T = \frac{1}{2}m\left(\dot{X}^2 + \dot{Y}^2\right)$$

$$\dot{X}^2 = \left[\dot{x}\cos(\omega t) - x\omega\sin(\omega t)\right]^2 + \left[\dot{y}\sin(\omega t) + y\omega\cos(\omega t)\right]^2 + 2\left[\dot{x}\cos(\omega t) - x\omega\sin(\omega t)\right]\left[\dot{y}\sin(\omega t) + y\omega\cos(\omega t)\right]$$

$$\dot{X}^2 = \left[\dot{x}^2\cos^2(\omega t) + x^2\omega^2\sin^2(\omega t) - 2\dot{x}x\omega\sin(\omega t)\cos(\omega t)\right] + \left[\dot{y}^2\sin^2(\omega t) + y^2\omega^2\cos^2(\omega t) + 2\dot{y}y\omega\sin(\omega t)\cos(\omega t)\right] + 2\left[\dot{x}\dot{y}\sin(\omega t)\cos(\omega t) + \dot{x}y\omega\cos^2(\omega t) - x\omega\dot{y}\sin^2(\omega t) - xy\omega^2\sin(\omega t)\cos(\omega t)\right]$$

$$\dot{X}^2 = \left[\dot{x}^2\cos^2(\omega t) + x^2\omega^2\sin^2(\omega t) - \dot{x}x\omega\sin(2\omega t)\right] + \left[\dot{y}^2\sin^2(\omega t) + y^2\omega^2\cos^2(\omega t) + \dot{y}y\omega\sin(2\omega t)\right] + \dot{x}\dot{y}\sin(2\omega t) + 2\dot{x}y\omega\cos^2(\omega t) - 2x\omega\dot{y}\sin^2(\omega t) - xy\omega^2\sin(2\omega t)$$

$$\dot{Y}^2 = \left[-\dot{x}\sin(\omega t) - x\omega\cos(\omega t)\right]^2 + \left[\dot{y}\cos(\omega t) - y\omega\sin(\omega t)\right]^2 + 2\left[-\dot{x}\sin(\omega t) - x\omega\cos(\omega t)\right]\left[\dot{y}\cos(\omega t) - y\omega\sin(\omega t)\right]$$

$$\dot{Y}^2 = \left[\dot{x}^2\sin^2(\omega t) + x^2\omega^2\cos^2(\omega t) + \dot{x}x\omega\sin(2\omega t)\right] + \left[\dot{y}^2\cos^2(\omega t) + y^2\omega^2\sin^2(\omega t) - \dot{y}y\omega\sin(2\omega t)\right] - \dot{x}\dot{y}\sin(2\omega t) + 2\dot{x}y\omega\sin^2(\omega t) - 2x\omega\dot{y}\cos^2(\omega t) + xy\omega^2\sin(2\omega t)$$

$$\mathcal{L} = \frac{1}{2}m\left(\dot{X}^2 + \dot{Y}^2\right)$$

$$\mathcal{L} = \frac{1}{2}m\left[\dot{x}^2\cos^2(\omega t) + x^2\omega^2\sin^2(\omega t) - \dot{x}x\omega\sin(2\omega t)\right] + \left[\dot{y}^2\sin^2(\omega t) + y^2\omega^2\cos^2(\omega t) + \dot{y}y\omega\sin(2\omega t)\right] + \dot{x}\dot{y}\sin(2\omega t) + 2\dot{x}y\omega\cos^2(\omega t) - 2x\omega\dot{y}\sin^2(\omega t) - xy\omega^2\sin(2\omega t)$$
$$+ \left[\dot{x}^2\sin^2(\omega t) + x^2\omega^2\cos^2(\omega t) + \dot{x}x\omega\sin(2\omega t)\right] + \left[\dot{y}^2\cos^2(\omega t) + y^2\omega^2\sin^2(\omega t) - \dot{y}y\omega\sin(2\omega t)\right] - \dot{x}\dot{y}\sin(2\omega t) + 2\dot{x}y\omega\sin^2(\omega t) - 2x\omega\dot{y}\cos^2(\omega t) + xy\omega^2\sin(2\omega t)$$

$$\mathcal{L} = \frac{1}{2}m\left[\dot{x}^2 + x^2\omega^2 + \dot{y}^2 + y^2\omega^2 + 2\dot{x}y\omega - 2x\dot{y}\omega\right]$$

Using Euler - Lagrange Eqn : $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}_i} = \frac{\partial \mathcal{L}}{\partial x_i}$

i) $m\left(\ddot{x} + \dot{y}\omega\right) = m\left(x\omega^2 - \dot{y}\omega\right)$

$$\ddot{x} + \dot{y}\omega = x\omega^2 - \dot{y}\omega$$

$$\boxed{\ddot{x} = \omega\left(x\omega - 2\dot{y}\right)}$$

ii) $m\left(\ddot{y} - \dot{x}\omega\right) = m\left(y\omega^2 + \dot{x}\omega\right)$

$$\boxed{\ddot{y} = \omega\left(y\omega + 2\dot{x}\right)}$$

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